

DATA ANALYTICS

CUSTOMER INSIGHTS

Customer Shopping Behavior Analysis

Analyzing 3,900 purchases across product categories to uncover spending patterns, customer segments, product preferences, and subscription behavior — guiding strategic business decisions.



DATASET

Dataset Summary

3,900

Rows

Transactional purchase records

18

Columns

Features per transaction

37

Missing Values

In Review Rating column

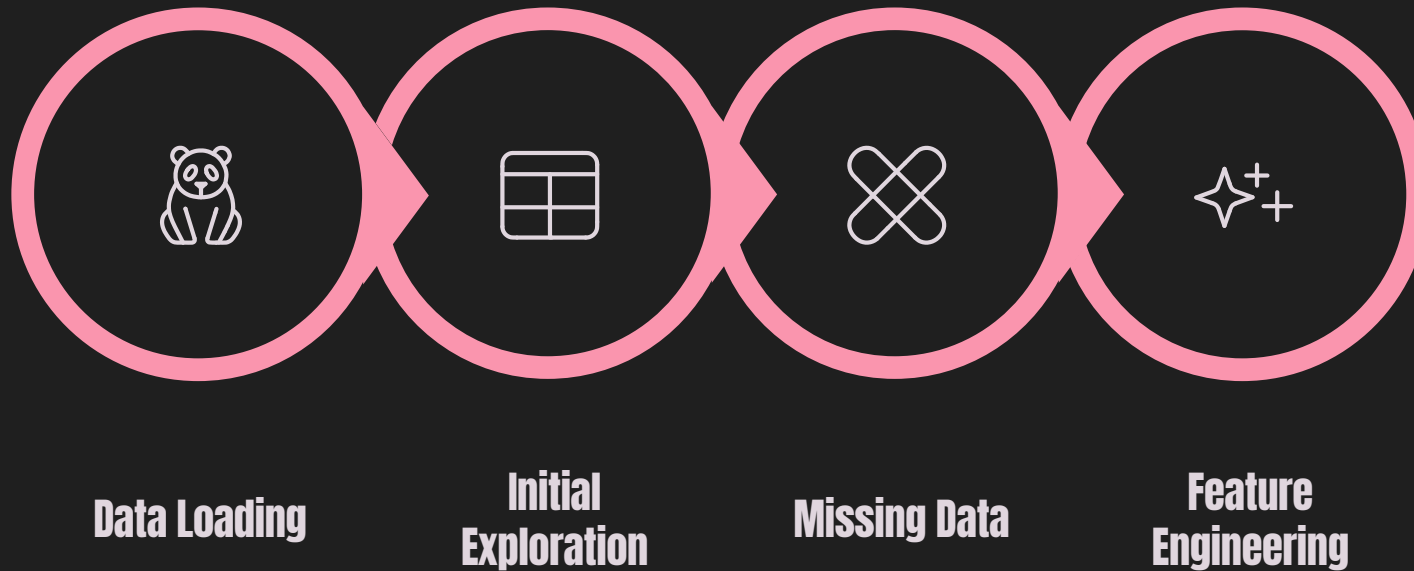
Key Features

- Customer demographics: Age, Gender, Location, Subscription Status
- Purchase details: Item, Category, Amount, Season, Size, Color
- Shopping behavior: Discount, Previous Purchases, Frequency, Rating, Shipping

EDA

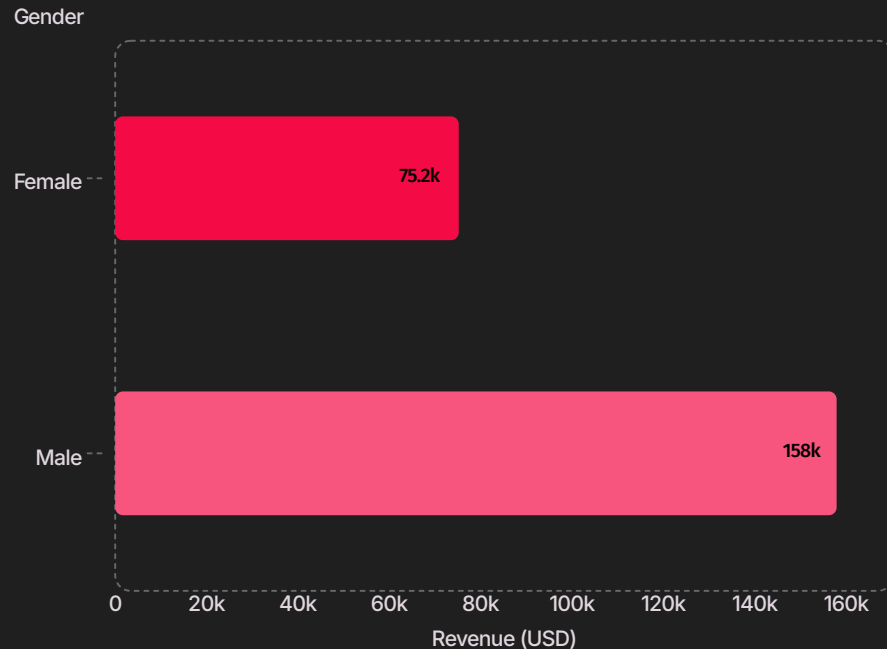
Exploratory Data Analysis in Python

Data preparation and cleaning pipeline using `pandas` for structured analysis.



Columns standardized to snake case; `promo_code_used` dropped as redundant with `discount_applied`. Cleaned data loaded into PostgreSQL for SQL analysis.

Revenue & Spending Insights



Male customers generated over **2× the revenue** of female customers.

High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount of \$59.76.

Shipping Type Comparison

Shipping	Avg Spend
Standard	\$58.46
Express	\$60.48

Product Performance Rankings

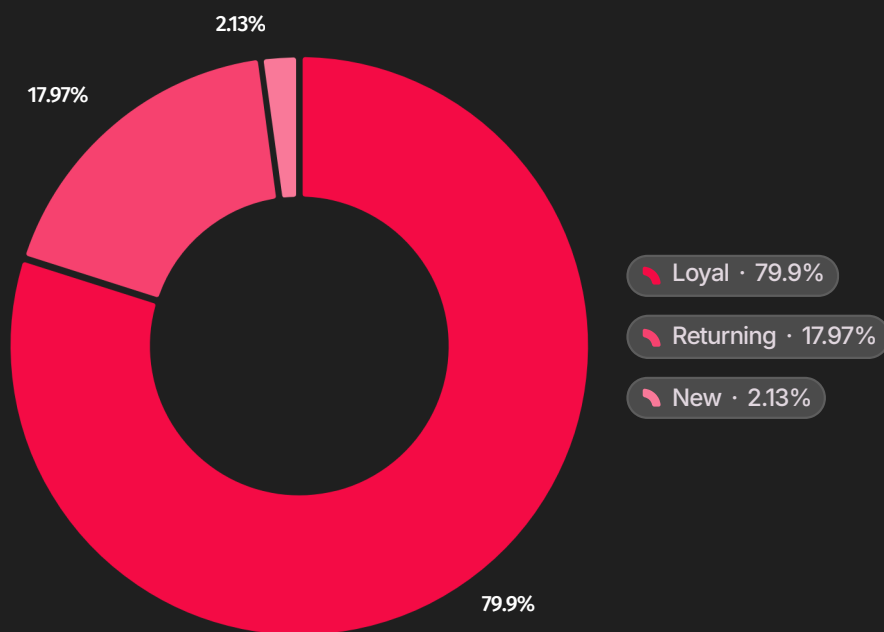
Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation



Repeat Buyers & Subscriptions

Among customers with more than 5 purchases:

- **2,518** are non-subscribers
- **958** are subscribers

Repeat buyers are significantly more likely to remain non-subscribers, suggesting an opportunity to convert loyal customers.

SQL ANALYSIS

Top Products per Category



Accessories

Jewelry (171), Sunglasses (161),
Belt (161)



Clothing

Blouse (171), Pants (171), Shirt
(169)



Footwear

Sandals (160), Shoes (150),
Sneakers (145)



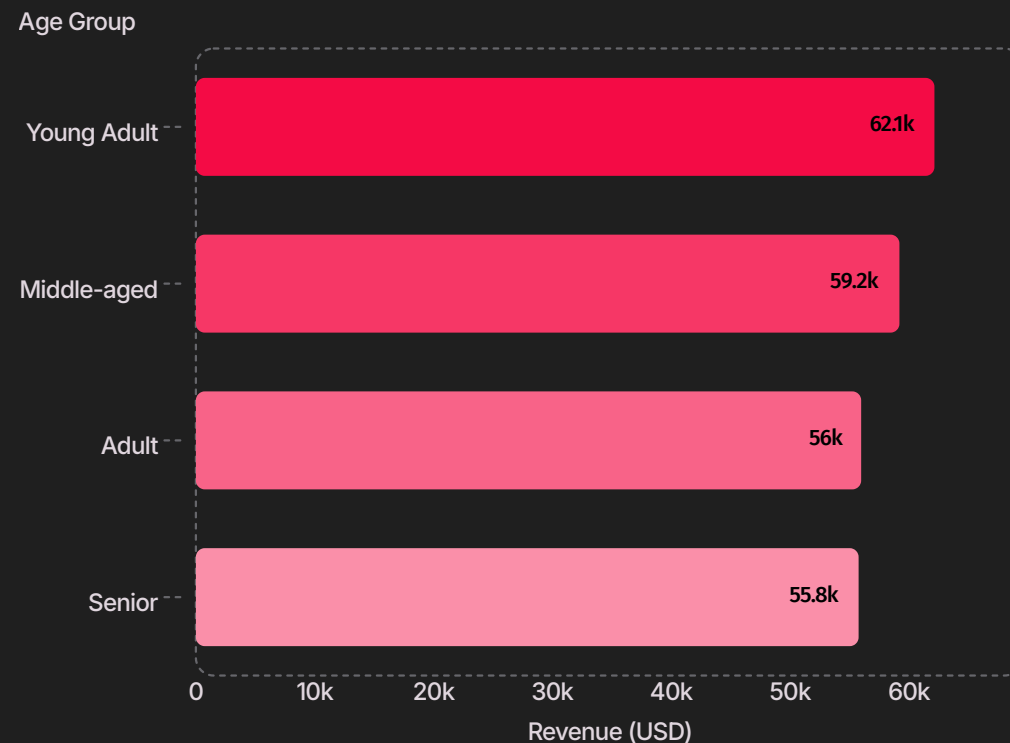
Outerwear

Jacket (163), Coat (161)

Subscribers vs. Non-Subscribers

Status	Customers	Avg Spend	Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Average spend is nearly identical across both groups, but non-subscribers drive **73% of total revenue** through volume.



Young Adults lead revenue contribution across all age groups.



POWER BI

Customer Behavior Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

27%

Subscribers

Interactive filters include Subscription Status, Gender, Category, and Shipping Type. Revenue by Category: Clothing leads at ~\$100K, followed by Accessories, Footwear, and Outerwear.

RECOMMENDATIONS

Strategic Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers to convert the 73% non-subscriber majority.

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control — 839 high-spenders still use discounts.

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.